

Problems Booklet

Dear participants,

The PLANCKS 2025 exam is finally in your hands!

Below we remind you some of the most important rules of the contest

- The language used in the competition is English.
- The contest consists of 9 problems. Problems 1-8 each are worth 10 points, problem 9 is worth 15 points. Subdivisions of points are indicated in the problems.
- Each problem must be handed separately, i.e. no piece of paper should contain the resolution of more than one problem. Please mark the header of each sheet with your team name, the problem number, and the problem page number. It is necessary to mark each sheet so that it is scored.
- When a problem is unclear, a participant may request clarification through the invigilator. If the response is relevant to all teams, the jury will provide this information to the other teams.
- Participants are allowed to use a dictionary: English to your native language.
- Participants are allowed to use a non-programmable, not-graph calculator (scientific is okay).
- No books or other sources, except for this Problem Booklet and a dictionary, are to be consulted during the competition.
- The use of hardware (including phones, tablets etc.) is not approved, except for watches and medical equipment. Please leave your phones in an envelope.
- Participants may not leave the exam room until one hour after the start of the competition. Afterwards, they must obtain permission from the invigilator to use the restroom.
- The jury has the right to disqualify teams for misbehavior or breaking the rules.
- In situations to which no rule applies, the Organizing Committee decides.

We hope you enjoy solving these problems. Good luck and may the best team win!

The Academic and Organizing Committees of PLANCKS 2025

UAB
Universitat Autònoma de Barcelona

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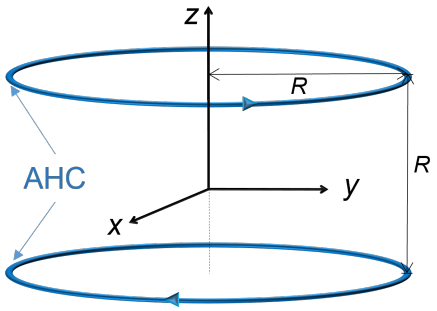
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1. Levitation in an Anti-Helmholtz coil

In some quantum magnetomechanical experiments a small magnetic particle is levitated in the magnetic field generated by two coaxial coils of the same radius R, separated by a distance R (from center to center). The same current, I, circulates in the coils in opposite directions for each coil. This system is called an anti-Helmholtz coil (AHC). The origin of the coordinates is located at the geometric center of the AHC system (see figure).



1. [**5 points**] Evaluate the magnetic induction field, $\mathbf{B}_a$, (in Cartesian coordinates) created by the AHC at points close to the origin, $|\mathbf{r}| \ll R$, up to the first non-vanishing order in the different coordinates of $\mathbf{r}$.

2. Imagine a small sphere of radius a ($a \ll R$) located at the origin of the coordinates. The sphere is made of linear, homogeneous, and isotropic (l.h.i) magnetic material characterized by magnetic susceptibility χ. The position of the sphere can be perturbed with small displacements, $\vec{\varepsilon}$ ($|\vec{\varepsilon}| \ll R$). The sphere is small enough so that the field inside it can be considered uniform with the value at its center and its magnetization distribution, $\mathbf{M}$, is also uniform. For the sphere, the demagnetizing field is $\mathbf{H}_d = -(1/3)\mathbf{M}$. Disregard gravity.

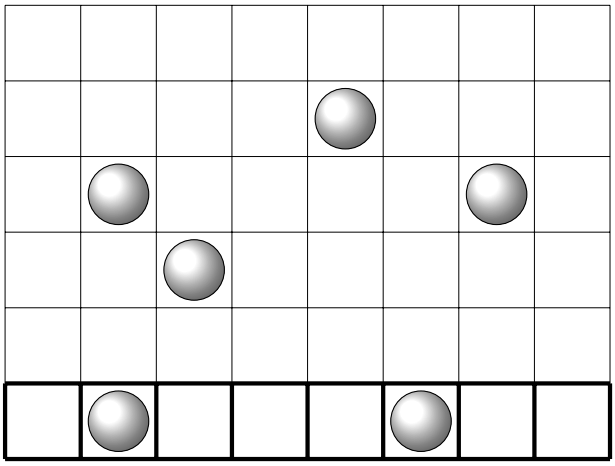
   a. [**2.5 points**] Consider $\chi = -1$. Calculate the magnetic force over the sphere, $\mathbf{F}$, at the origin and study the stability of the system after small displacements, in terms of the stiffness coefficients, $\kappa_{\alpha\beta}$. The stiffness coefficients are defined as

   $$\kappa_{\alpha\beta} \equiv -\left.\frac{\partial F_\alpha}{\partial \beta}\right|_{\mathbf{r}=0},$$

   where $\alpha, \beta = (x, y, z)$.

   b. [**2.5 points**] Consider $\chi = +1$. Calculate the magnetic force over the sphere, $\mathbf{F}$, at the origin and study the stability of the system after small displacements, in terms of the stiffness coefficients, which are defined as before.

2. Surface Adsorption



Gas adsorption on solid surfaces has a significant impact on the design of advanced materials for many applications. For instance, materials such as MOFs (Metal-Organic Frameworks) are actively studied to capture CO_2 or hydrogen, of great interest in environmental science and energy storage. In this problem, we will explore a simple model of gas adsorption on a surface.

Let us consider a gas confined in a container and study the adsorption of particles at the walls. We will study the process by neglecting the kinetic energy of the particles and using a discrete model for the locations of the particles in the bulk and on the surface (see figure). Let N_B be the number of possible spatial cells the particles may occupy in the bulk and N_S be the number of spatial cells on the surface. The gas consists of N particles and $N_B \geq N \geq N_S$. Let n be the number of particles actually on the surface: $0 < n \leq N_S$. A particle has an energy $-\epsilon$ (binding energy) while it is on the surface and 0 while it is in the bulk. N_B, N_S and N are all constants and very large; n is a variable and also large. The volume of a gas cell is V_0.

1. [**2 points**] Using the microcanonical ensemble find the entropy as a function of n. Make use of the Stirling approximation.

2. [**2 points**] Derive an expression relating n to the temperature of the system. Find the limit of n/N when $T \to \infty$ and $T = 0$ reasoning your results.

3. [**2 points**] Derive the expression for $n = n(T)$ but now using the canonical ensemble. Compare your result with that derived in question (2) and reason your results.

4. [**2 points**] Assume $N_B \gg N \gg N_S$. Find the explicit relation $n = n(T)$. For which temperature value a surface cell is equally likely to be occupied or unoccupied?

5. [**2 points**] In the limit $N_B \gg N \gg N_S$, find the fraction of adsorbed particles $\theta \equiv n/N_S$ in terms of the gas pressure and plot θ vs p for a fixed temperature (this is named Langmuir isotherm). Find the pressure for which half of the particles are adsorbed. Give a physical interpretation of the dependence of θ with p.

3. Second sound

Heat transport in semiconductors is essential in many areas of nanotechnology. For instance, the rate at which heat is evacuated in nanocircuits determines the actual limits for the speed of computation in what is called the "thermal wall". In bulk materials, heat transport is usually described by Fourier's law, and it is said to be diffusive. Things work differently when the system length or time scales are comparable to the mean free path or mean free times of heat carriers. Here we will study the possibility of heat waves, what is known as second sound. It was first observed in the 70's at cryogenic temperatures for just a few materials, and recently in graphite and in germanium at larger temperatures. To this end, we consider the so-called Maxwell-Cattaneo equation for the heat flux $\mathbf{q}$

$$\tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -\kappa \nabla T, \tag{1}$$

with T the temperature, κ the thermal conductivity and τ the heat flux relaxation time, which in semiconductors is an average mean free time for phonon resistive scattering.

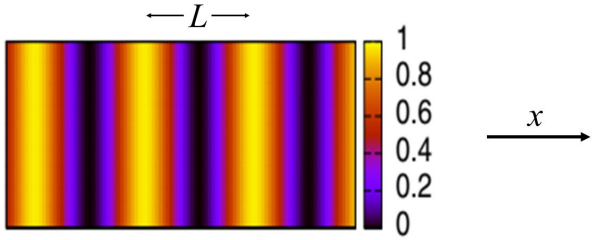
In what follows, assume the Maxwell-Cattaneo equation and that volume does not change.

Questions

1. [**1 point**] Show that in some limit one recovers Fourier's law.

2. [**2 points**] Show that in some limit the temperature propagates as a wave, and that the speed of this wave is smaller than (first) sound speed. Hint: consider that the thermal conductivity is $\kappa = \frac{1}{3} C_V c^2 \tau$, with C_V the heat capacity per unit volume and c the speed of sound; this expression can be obtained at low enough temperatures for three dimensional materials assuming Debye model with identical branches.

One type of experiment to detect the wave-like behaviour of temperature is by instantaneously heating the surface of the dielectric with a laser with a spatially periodic pulse of period L, providing an initial perturbation $\Delta T(x,0) = A\cos(2\pi x/L)$ (see figure), and measure how the system relaxes to equilibrium (these are called thermal grating experiments). Let us assume that heat flows only in the x axis.



3. [**3 points**] Find the evolution of the induced temperature difference $\Delta T(x,t)$ and the range of L for which one could observe at least one temperature oscillation before reaching equilibrium.

4. [**1 point**] Discuss the possibility of finding that heat flows from cold to hot.

5. [**3 points**] Let us now analyze the compatibility of the Maxwell-Cattaneo equation (1) with the second law of thermodynamics. Find the entropy production assuming local-equilibrium and discuss its sign. Do the same if one assumes a non-equilibrium entropy density of the type

$$s(u,\mathbf{q}) = s_{eq}(u) - \frac{\tau}{2\kappa T^2}\mathbf{q}^2,$$

where u is the energy density and s_{eq} the local-equilibrium entropy density.

Note: neglect terms higher than quadratic in the entropy production.

4. Some questions about our universe

• Answer the questions in a clear, structured and concise manner, underlining the final result.
• Before starting to solve the problems, read the Formulae section on the next page, where you will find some useful equations and definitions that you need.

1.a. [**1 point**] Assume a component of the universe A having an equation of state relating pressure and energy density of the form $P_A = w_A\,\rho_A$, with constant w_A. Use the Friedmann eqs in the Formulae to show that the behaviour of ρ_A with respect the scale factor a is of the form

$$\rho_A \propto a^{\pi_A}$$

and calculate π_A as a function of w_A.

1.b. [**1 point**] The dependence of the energy density on a can be deduced from physical arguments in the case of $A = M, R, \Lambda$. (i) For matter, $\rho_M \propto a^{-3}$ because of dilution due the expansion of the universe, (ii) for radiation, $\rho_R \propto a^{-4}$ because in addition of dilution we have energy redshift, and (iii) for the cosmological constant, the energy density ρ_Λ is constant (independent of a).

Use the facts (i), (ii) and (iii) and what you got in **(a)** to deduce the values of w_M, w_R, w_Λ.

(FYI: The form of $\rho_M(a)$, $\rho_R(a)$, and $\rho_\Lambda(a)$ allows to catch some properties of the evolution of the universe.)

2.a. [**2 points**] **(a)** From the Friedmann eqs shown in Formulae one can find an equation giving $\ddot{a}(t)$

$$\ddot{a} = aG\,(\alpha\rho + \beta P)$$

(ρ = total energy density, P = total pressure). Calculate the value of the constants α and β.

2.b. [**2 points**] Define the deceleration parameter q_0

$$q_0 = -\left.\frac{a\ddot{a}}{\dot{a}^2}\right|_0$$

where $a, \dot{a}$ and $\ddot{a}$ are taken at $t = t_0$ (today's time).

Calculate q_0 as a function of Ω_M, Ω_R and Ω_Λ (and nothing else).

(FYI: In our universe Ω_Λ dominates, leading to $q_0 < 0$, i.e. now the universe is accelerating)

3. Consider a flat universe ($k = 0$) with matter and cosmological constant ($\rho_R = 0$). The objective is to calculate the age of such an universe t_0.

   a. [**2 points**] Show that t_0 is given by an integral

   $$t_0 = \int_{\alpha_1}^{\alpha_2} da\, f(a, H_0, \Omega_M, \Omega_\Lambda)$$

   Find the values of integral limits α_1 and α_2. Find the function f.

   b. [**2 points**] Solve the integral, and find t_0 as a analytical function that depends only on H_0 and on Ω_Λ. (You will need several substitutions, you might start with $x = a^{3/2}$)

(FYI: With the observed values of H_0 and Ω_Λ we get the age of our universe, to a very good approximation.)

Formulae

We denote by $a = a(t)$ the scale factor of the universe (we will normalize $a(t_0) = 1$) and by $H = H(t) = \dot{a}/a$ the Hubble parameter (function of time t), and the Hubble constant by $H_0 = H(t_0)$. The time t_0 is today's time. As usual $\dot{a} = da/dt$ and $\ddot{a} = d^2a/dt^2$.

The total energy density of the universe is given by the sum of all contributions $\rho = \sum_A \rho_A$ and similarly for the total pressure, $P = \sum_A P_A$. The first and second Friedmann equations are

$$\text{1st eq:} \qquad H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\,\rho - \frac{k}{a^2} \qquad \text{2nd eq:} \qquad \frac{d\rho_A}{dt} = -\,3H\,(\rho_A + P_A)$$

(G = Newton's const., k =curvature of the universe). Note that the 2nd eq is actually an equation for each A. Summing over A we get $d\rho/dt = -\,3H\,(\rho + P)$.

We define the critical density ρ_c and the quantities Ω_A

$$\rho_c = \frac{3H_0^2}{8\pi G}\,, \qquad \Omega_A \equiv \frac{\rho_A(t_0)}{\rho_c}$$

In the problem that we propose, we consider 3 (ideal) fluid components in the universe: matter (M), radiation (R), and cosmological constant (Λ). (M is non relativistic matter and R are photons or similar.) In these cases, the relation between energy density and pressure (equation of state) is linear: $P_A = w_A\,\rho_A$, with $w_M = 0$, $w_R = 1/3$, $w_\Lambda = -1$. Finally, we have $\Omega_M = \rho_M(t_0)/\rho_c$, and similarly for R and Λ.

5. Black Hole Shadow

We would like to observe the "shadow" of a black hole with an earth-based telescope system. To do so, we consider a Schwarzschild black hole around which matter orbits at radii ranging from the innermost stable circular orbit (ISCO) until larger distances. That matter will emit light, which we want to picture with our telescopes. Note: In this exercise we do not consider bending of light rays in the vicinity of the black hole.

1. In order to calculate the radius of the ISCO, we consider the effective potential energy given by General Relativity (GR) for a test mass m, located at distance r from the black hole center and angular momentum L:

$$V_{\text{eff}}(r) = -\frac{GmM}{r} + \frac{L^2}{2mr^2} - \frac{R_s}{2} \cdot \frac{L^2}{mr^3}, \tag{1}$$

where G is the gravitational constant, M the mass of the black hole, $R_s = 2\,GM/c^2$ the Schwarzschild radius and c the light speed.

   a. [**1.5 points**] Find the effective potential energy for Newton's law of gravitation in terms of the angular momentum and compare with the GR one. In which situations the GR effects will be relevant and in which not?

From now on consider Eq. (1) and that L is a constant of motion.

   b. [**1 point**] Find the orbits $r = r(L)$ and show that there exists a minimum value of L.

   c. [**4 points**] Find the range of possible orbits r, the range of r for stable solutions and the ISCO radius. Hint: you may find helpful to plot L^2 vs r, but it is not mandatory to reach the maximum mark.

2. A relatively close and active supermassive black hole in the vicinity of our Galaxy has a mass of approximately $5 \times 10^9 M_\odot$ (where the solar mass $M_\odot = 2 \times 10^{30}$ kg), and a distance of 16.7 Mpc (1 pc = 3.1×10^{16} m).

   a. [**0.5 points**] Calculate the angular extension of the ISCO radius of that supermassive black hole, if observed from Earth.

   b. [**3 points**] To picture that black hole, we need to design a system that reaches an angular resolution better than the angular extension of the object derived in the previous section, in the best case at least twice as good. The best angular resolutions can be achieved through interferometric observations, where the same incident wave is detected by pairs of distant telescopes, and their signals get multiplied by each other to generate an interference pattern (in the so-called correlator). The angular resolution is then approximately λ/D, where λ denotes the wavelength of the observed light wave and the D the distance between a telescope pair. At large distances, both telescopes can only register the signals, digitize them and correlate them later offline. Technically, delay precisions between different stations of the order to ~ 1 ps are feasible... (continued on the next page)...

The following picture shows the transparency of the terrestrial atmosphere to light at different wavelengths. Can you choose a suitable observation wavelength for the interferometer and the required minimum distance between two telescope pairs to achieve the angular resolutions found in question 2a?

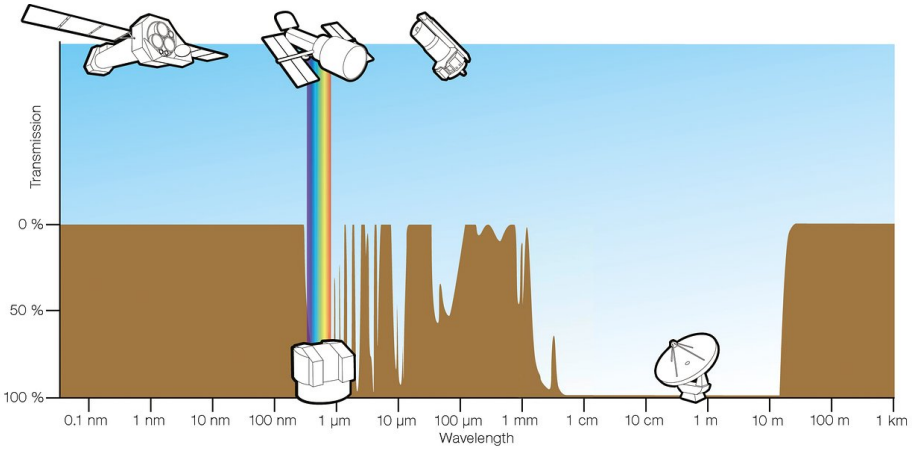
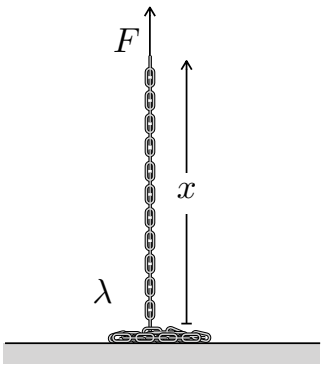


Figure 1: Transparency of the Earth's atmosphere. Note the inverted vertical scale. Credit: ESA/Hubble (F. Granato).

Constants: $G = 6.7 \times 10^{-11} \mathrm{m}^3\,\mathrm{kg}^{-1}\mathrm{s}^{-2}$; $c = 3 \times 10^8\,\mathrm{m/s}$; Earth radius= $6378\,\mathrm{km}$.

6. Pulling a chain upwards with constant force

Consider pulling one end of a very long chain initially piled up on a table with a constant force F, as illustrated in the figure. At the start, a very short segment of length x_0 is held vertically at rest. The chain is subject to Earth's gravitational attraction, with a constant acceleration g.
The chain consists of links that are infinitesimally small in both length and mass, allowing it to be treated as a continuous system with a uniform linear mass density λ. Neglect the height and horizontal spread of the pile on the table.



1. [**3 points**] Derive the ordinary differential equation for the velocity v of the rising segment of a chain as a function of its height x (1 point), and solve it to obtain the expression (2 points)

$$v = \sqrt{\frac{F}{\lambda}\left(1 - \frac{x_0^2}{x^2}\right) - \frac{2}{3}gx_0\left(\frac{x}{x_0} - \frac{x_0^2}{x^2}\right)}$$

   Hint: You may need an integrating factor $\mu(x)$ to solve the equation.

2. [**1 point**] Determine the maximum height $x_{\max}$ of the chain that can be reached with the applied force.

3. [**1 point**] Explain why $x_{\max}$ exceeds the equilibrium height $x_{\text{eq}} = F/(\lambda g)$, where the force F balances the chain's weight.

In the remaining sections, assume that the initial height x_0, where pulling begins, is much smaller than the equilibrium height, i.e., $x_0 \ll x_{\text{eq}}$.

4. [**1 point**] Determine $x(t)$ for timescales much larger than $x_0\sqrt{\lambda/F}$ and up to the moment when the end of the chain reaches its maximum height.

5. [**2 points**] Obtain the velocity of the descending segment of the chain as a function of x after the chain has reached its maximum height $x_{\max}$.

6. [**1 point**] Calculate the minimum height, $x_{\min}$, that the vertical segment of the chain will reach during its descent.

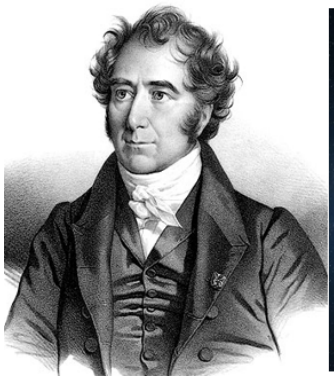
   Hint: The equation

$$\frac{3}{2} - \xi + \log\left(\frac{2\xi}{3}\right) = 0$$

   has two solutions, $\xi_0 = 3/2$ and $\xi_1 \approx 0.625$.

7. [**1 point**] Sketch an approximate graph of $x(t)$ for a time interval that includes several cycles as described in the previous questions.

7. Stellar observation: from diffraction to interferometry





In 1818, Augustin-Jean Fresnel presented his work on the wave theory of light in front of the French Academy of Sciences. The panel included eminent scientists of the time, such as Simèon Denis Poisson and François Arago. Fresnel argued that light behaved as a wave, contrary to Newton's corpuscular theory, which still had many followers.

During the presentation, Poisson, a staunch supporter of the corpuscular theory, argued that if Fresnel's wave theory were correct, a bright spot should be observed at the centre of the shadow cast by an opaque circular object, which seemed absurd. However, Arago decided to test this prediction. To the surprise of many, the experiment confirmed the existence of this bright spot, now known as Poisson's spot or Arago's spot. This result provided strong evidence in favour of the wave theory of light, marking a milestone in the history of optics.

Many years have passed since Arago conducted his famous experiment, and in that time, humanity has achieved incredible feats, such as sending telescopes into space that bring us closer to the deepest mysteries of the universe. However, what Fresnel and his colleagues discovered remains vitally important today. Their work not only revolutionized our understanding of light but also laid the foundation for the development of advanced optical technologies. Today, the diffraction phenomenon they studied is fundamental to understanding the resolving power of optical instruments like the James Webb Space Telescope, allowing us to explore the cosmos with unprecedented clarity.

Questions

Suppose Arago conducts a second experiment where he illuminates an opaque square object, with side length l, using coherent light (both spatially and temporally) emitted by an oil lamp with a central wavelength λ_c. The light source is placed at a sufficiently large distance from the object to consider that it is illuminated with plane waves.

1. [**3 points**] Applying the Fraunhofer approximation, calculate the intensity distribution (diffraction pattern) as a function of the position x and y on the screen if it is located at a distance $D \gg l$, and the angular position of the first diffraction minimum in direction x and direction y. Is there a bright spot?

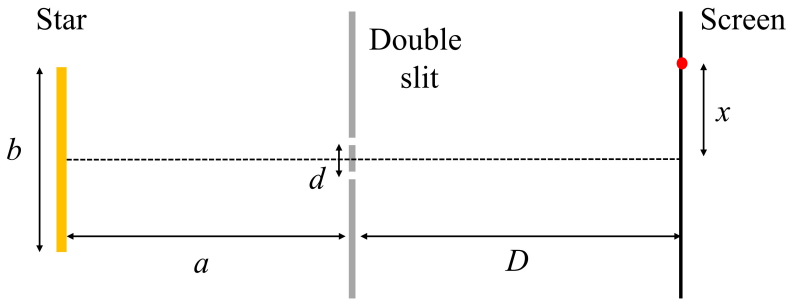
2. Let us now assume an astronomical telescope with a square aperture of side length L:

   a. [**1 point**] Write the equation for the angular resolving power of this telescope using Rayleigh's criterion. Justify your answer.

   b. [**1 point**] Determine if this telescope can resolve as separate objects the binary star system of Sirius, composed of Sirius A and Sirius B, with an angular distance of approximately $\theta = 10$ arc seconds. Note: Assume that Sirius A and B present the same energy flux at the telescope plane, that we are observing in the visible range with an average wavelength of $\lambda = 550$ nm. The telescope's square aperture has a side length of $L = 2.4$ meters. (Note: $1^\circ = 3600$ arc seconds).

3. [**3 points**] Now we move the telescope to observe the Orion constellation. We set up a double slit (Young's double slit) attached to the telescope objective to implement a Michelson stellar interferometer (see the figure below for reference). Let us consider the one-dimensional model depicted in the figure, where b is the size of a star, d the distance between slits, a is the distance of the star to the telescope, and D the distance of the slits

to the screen ($a, D \gg d$). Assuming that the intensity emitted by the star is uniform, calculate the visibility function, $V = \frac{I_{max}-I_{min}}{I_{max}+I_{min}}$. (Note: $\sin A - \sin B = 2\sin\frac{A-B}{2}\cos\frac{A+B}{2}$). Neglect the diffraction effects of the aperture.



4. [**2 points**] When Michelson used the stellar interferometer to study the star Betelgeuse, a red giant star located in the Orion constellation, he found that the first minimum of visibility of the interference fringes occurred at a distance between the slits of d = 308 cm. Using an average wavelength of 570 nm, what is the angular diameter of the star? If the distance to Orion is estimated to be 1.6 10^{15} km, what is the diameter of the star?

8. Graphene nanoribbons

Introduction

Graphene lacks an electronic band gap, limiting its application in electronic devices. However, graphene nanoribbons (Figure 2 (left)) exhibits different behavior. Let's estimate the band gap opening in graphene nanoribbons using an approximate energy dispersion evaluation as a function of the wave vector k. A suitable method to calculate the energy bands is the tight-binding approximation (TBA), which approximates the electronic wave function as a linear combination of atomic orbitals. The lattice periodicity is introduced via Bloch's theorem $\psi_k(x+na) = e^{ikna}\,\psi_k(x)$ where a is the translation vector in 1D, and n is an integer. In this exercise, we will restrict to interactions between nearest neighbors. As an example of TBA in a one-dimensional chain, consider an infinite polyacetylene carbon chain with single and double bonds (Figure 1(left)).

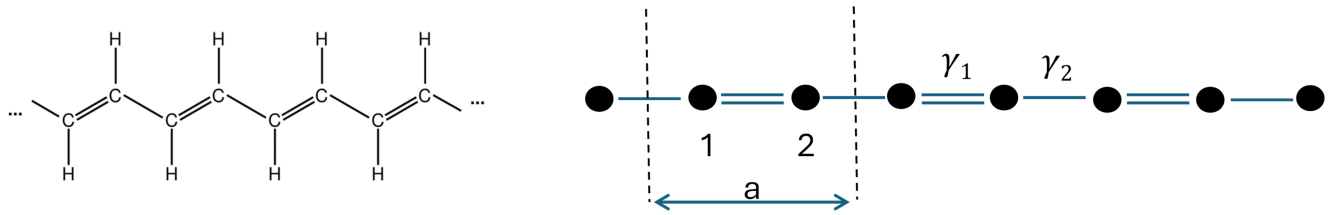


Figure 1: (left) Polyacetylene chain; (right) schematic representation

The unit cell in this case is composed of two atoms, labeled atom 1 and atom 2 (fig. 1 (right)). Within TBA the wavefunction can be expressed as a linear combination of the wavefunctions in the unit cell, $\psi^i(x)$, which can be written in terms of the atomic orbital wavefunctions $\phi_i(x)$, each weighted by a phase factor (Bloch's theorem):

$$\psi_k(x) = \zeta_1\psi_k^1(x) + \zeta_2\psi_k^2(x) \quad \text{with} \quad \psi_k^{1,2}(x) = \frac{1}{\sqrt{N_c}}\sum_{n=1}^{N_c} e^{ik(x+na)}\,\phi_{1,2}(x+na) \tag{1}$$

where n runs over the N_c unit cells. To solve the Schrödinger equation, we project it onto $\langle\psi_k^1|$ and $\langle\psi_k^2|$, resulting in a system of two equations. We assume that only nearest neighbor interactions are non-zero. The hopping integral $\langle\phi_i|H|\phi_j\rangle$ for the interaction between $1 \to 2$ is γ_1 and for the interaction between $2 \to 1$ is γ_2 (fig. 1b). The on-site energy, $\langle\phi_i|H|\phi_i\rangle = \varepsilon_0$ and the orbital overlap term $\langle\phi_i|\phi_j\rangle = \delta_{ij}$. This yields the eigenvalue equation (Eq. 2) that needs to be solved for the energy, ε_k. In summary, each term of the matrix accounts for interactions between i and j, as shown by the rows and column labels, weighted by phase factors such as $e^{\pm ika}$, when interacting with nearest neighbors in different cells:

$$\begin{array}{c} \\ 1 \\ 2 \end{array}\!\!\begin{array}{c} \begin{array}{cc} 1 & \quad 2 \end{array} \\ \begin{bmatrix} \varepsilon_0 & \gamma_1 + \gamma_2\, e^{-ika} \\ \gamma_1 + \gamma_2\, e^{+ika} & \varepsilon_0 \end{bmatrix} \end{array} \begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix} = \varepsilon_k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix} \tag{2}$$

Questions

Armchair graphene nanoribbons are one-dimensional, infinitely long linear chains with a width W, as illustrated in Figure 2 (left).

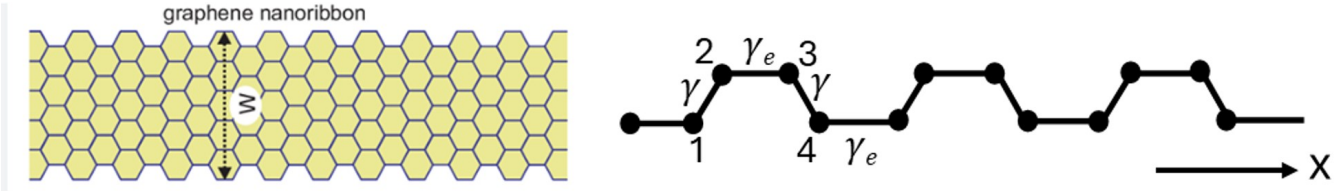


Figure:2 (left) armchair graphene nanoribbon; (right) schematic representation used for the TBA in this exercise

Our goal is to use tight-binding approximation, as explained in the introduction, to calculate the energy dispersion relation. As a simplification take the simplest infinite nanoribbon along direction x, as shown in fig. 2 (right). Assume the on-site energy, ε_0, is zero, and the hopping integral between nearest neighbors are γ and γ_e (fig. 2 (right)), with values $\gamma = -2.70$ eV and $\gamma_e = -3.02$ eV.

1. [**1 point**] Construct the unit cell.
2. [**1 point**] Write the wave function of an electronic orbital in the unit cell.
3. [**2.5 points**] Build the matrix of the eigenvalue equation, either by applying TBA or by similarity with Eq. (2).
4. [**2 points**] Calculate the energy gap at the Γ point ($k = 0$).
5. [**1 point**] What would happen with the energy gap if the interaction between atoms in the nanoribbon is the same for all atoms, i.e. $\gamma = \gamma_e = -2.70$ eV?
6. [**2.5 points**] Make a schematic representation of: i) energy E versus wave vector k between 0 and π/a for all the bands; ii) E versus the density of states (DOS) for the valence band. Note: Recall that the DOS in 3D can be evaluated according to $g(E) = \frac{2}{(2\pi)^3} \int_S \frac{dS}{|\nabla_k E|}$, where the integral is performed on a constant energy surface, and the factor 2 accounts for spin degeneration.

9. Quantum Grover algorithm and classical collisions

The **quantum Grover algorithm** is an oracle algorithm that can find a register in an unstructured list of N registers (Imagine finding a name in a telephone book if we are given the telephone number).
The idea is to start with an equally weighted state of all registers:

$$|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle,$$

and do some operations that enhance the amplitude of the marked register denoted by $|w\rangle$. One first writes $|\psi_0\rangle$ in a two-dimensional basis

$$|\psi_0\rangle = \sqrt{\frac{N-1}{N}}|\alpha\rangle + \sqrt{\frac{1}{N}}|w\rangle,$$

where $|\alpha\rangle$ is the state of all unmarked registers (of course, $\langle\alpha|w\rangle = 0$). In the $\{|\alpha\rangle, |w\rangle\}$ basis, the oracle is given by a unitary matrix

$$U = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

that adds a phase, -1, to the amplitude of the marked state $|w\rangle$. The Grover algorithm is the following:

(I) Apply the oracle U

(II) Do a unitary operation that can be seen as a reflection over $|\psi_0\rangle$

$$K = 2|\psi_0\rangle\langle\psi_0| - \mathbb{1},$$

(III) Repeat operations (I) and (II) until the amplitude of $|w\rangle$ is maximal, i.e. when measuring the state after these iterations one obtains $|w\rangle$ with maximum probability.

1. [**2 points**] Show that KU can be written as a rotation matrix $G(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, and find the θ as a function of N.

2. [**2 points**] Show that if $N = 4$, one iteration of the Grover algorithm is sufficient to find the marked state. [Recall that $\cos\theta/2 = \sqrt{(1+\cos\theta)/2}$ and $\sin\theta/2 = \sqrt{(1-\cos\theta)/2}$].

3. [**2 points**] Find the general expression of N (although in general, it will not be a natural number) in terms of the number of times, n, the oracle must be called for the Grover algorithm to find the exact marked state. For solutions of large N, use the Taylor expansion to find the asymptotic expression of n as a function of N.

It is interesting to realize that the quantum Grover algorithm is formally equivalent to the following **classical scattering problem**. Imagine a dystopian classical world where energy has become very scarce. In this scenario, we aim to design a mountain train that operates without consuming energy, relying solely on the physics of elastic collisions.

The train consists of two parts: a locomotive without an engine of mass m_1 and a coach of mass m_2, with $m_1 > m_2$. The locomotive is positioned at the rear, while the coach leads at the front. Starting from rest at an initial height on a mountain, the train descends toward a bottom station, reaching a final velocity $v = 1$ (in arbitrary units).

Although the train's components descend simultaneously, they are not physically connected. Upon reaching the bottom station, the coach collides with a wall, reversing its velocity, and subsequently collides with the locomotive. These collisions may be repeated multiple times. All collisions are assumed to be perfectly elastic and to occur sequentially.

4. [**2 points**] Demonstrate that this classical scattering problem is formally equivalent to the operations of the Grover algorithm, where the oracle U corresponds to the collision of m_2 with the bottom wall and K represents the locomotive-coach collisions. Derive the expressions relating m_1 and m_2 to N.

   Hint: The following rescaling of the velocities is very useful: $\xi_i = \sqrt{m_i/M}\, v_i$ where $M = m_1 + m_2$. To ease the notation, it is also convenient to define the ratio $\eta = m_1/m_2$.

5. [**2 points**] What is the minimum mass ratio, $\eta = m_1/m_2$ (with $\eta > 1$), required for the coach to remain stopped at the bottom station after the train descends?

6. [**1 point**] If the train took a time T to descend, how much time do passengers have to do their shopping before the locomotive returns to the bottom station after reaching its maximum height? (assume the mountain is a sloped plane)

7. [**2 points**] In the above scenario, when the locomotive reaches the bottom station it collides with the stopped coach and a second round of collisions occurs. When this round ends, what are the final velocities of the coach and the locomotive?

8. [**2 points**] Derive the general expression for the mass ratio η under which this "park-and-pick-up" phenomenon occurs.